Sports Medicine
https://doi.org/10.1007/s40279-020-01407-4

SYSTEMATIC REVIEW

Detrimental Effects of the Off-Season in Soccer Players: A Systematic Review and Meta-analysis

Filipe Manuel Clemente[1,2] · Rodrigo Ramirez-Campillo[3,4] · Hugo Sarmento[5]

Accepted: 30 November 2020
© The Author(s), under exclusive licence to Springer Nature Switzerland AG part of Springer Nature 2020

Abstract

Background The off-season period in soccer leads necessarily to changes in fitness status. However, there is a lack of systematization that allows identifying the magnitude of these changes in groups participating in off-season training programs compared with those subjected to training cessation.

Objective This systematic review with meta-analysis was conducted to assess the effects of training cessation in off-season training programs on men soccer players' body fat, maximal oxygen uptake (VO_{2max}), yo–yo intermittent recovery test (YYIRT), vertical jump, sprinting time, and repeated-sprint ability.

Methods To qualify for inclusion in the systematic review, studies must have included: (1) a detraining period of ≥ 2 weeks; (2) controlled trials or cohorts of healthy men soccer players with no restriction on age; and (3) a pre–post training cessation or off-season training programs measure of body fat (%), VO_{2max} (mL kg^{-1} min^{-1}), YYIRT performance (meters), vertical jump (height), sprinting (time), and repeated-sprint ability (total time).

Results The electronic search yielded 563 articles, and 12 were subsequently included. Significant (all $p < 0.05$) detrimental training cessation effects were noted for body fat (ES = 0.26), VO_{2max} (ES = − 1.48), YYIRT (ES = − 0.46), vertical jump (ES = − 0.81), and repeated-sprint ability (ES = 0.68). Similarly, significant (all $p < 0.05$) detrimental off-season training programs effects were noted for body fat (ES = 0.26), VO_{2max} (ES = − 0.48), vertical jump (ES = − 0.51), and sprinting time (ES = 0.86). When training cessation and off-season training programs effects were compared, greater detrimental effects were noted after training cessation for VO_{2max} ($p = 0.002$) and repeated-sprint ability ($p < 0.001$).

Conclusions Detrimental effects on body composition and physical fitness were observed after both training cessation and off-season training programs. However, off-season training programs seem to ameliorate such detrimental effects on VO_{2max} and repeated-sprint ability to some extent. The results presented here call for the implementation of more effective off-season training programs among male soccer players.

1 Introduction

Pre-season, in-season, and off-season (or transition) periods are the common periods of a full soccer season [1]. The pre-season phase in professional sports is typically a brief period in which an intensification of training load (mainly, external load—physical demands) occurs in comparison to typical in-season weeks [2, 3]. Comparisons of these phases in youth athletes are not very well-described

Key Points

- Detrimental effects on body composition and physical fitness occur after training cessation and off-season training programs.
- Off-season training programs seem to offer some value in ameliorating the detrimental effects on VO_{2max} and repeated sprint ability.
- The introduction of specific training elements (e.g., strength/power, sprinting) is recommended as part of off-season training programs to sustain a well-prepared fitness status.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40279-020-01407-4.

✉ Filipe Manuel Clemente
filipe.clemente5@gmail.com

Extended author information available on the last page of the article

Published online: 05 January 2021

in the literature. Usually, the intensification during the pre-season is intended to stress the body and promote positive adaptations in the fitness status of players, as has been suggested in studies conducted in professionals [4, 5] and young players [6]. The off-season is often characterized by a decrease in training load, a complete cessation of training, or a transition to individualized exercise at low-to-moderate levels during interruptions in competitions [1, 7]. Meaningful decreases in training volume, intensity, and frequency during the off-season can lead to detraining (DT) and have possible detrimental effects on performance and body composition [8, 9]. Of note, DT can occur not only during the off-season, but also during in-season breaks (winter break) [10].

Although the off-season period varies from country to country, it usually lasts 4–6 weeks [11, 12], while an in-season break might be 2 weeks long [10]. Such periods of training cessation or reduced volume can be categorized as long term (> 4 weeks) or short term (< 4 weeks) DT [7]. Both categories induce reversibility of fitness levels and body composition due to the decrease in training stimuli. It has been reported that DT conduct decreases players' endurance performance, leads to small-to-moderate negative changes in body composition (loss of lean mass and gain of body fat, moderate increases in sprinting time (ST), small-to-moderate decreases in muscle power, and a reduced capacity to produce force at high angular velocity [1]. Owing to the length of the off-season, players regularly split the off-season between a period of training cessation (in which no planned program of training is performed) and a period of training restart (which involves individualized training programs) [8, 9]. The use of an off-season training program (OTP) is intended to mitigate losses in athletic performance [13] and to control negative changes in body composition [14]. Therefore, the off-season should be thought of as a transition period; thus, part of the period should be used to restart the training process and, in turn, to progressively adjust the player's fitness level to the necessary status to face and cope with the load imposed during the pre-season [1].

Moreover, DT strategies that can prevent/modulate reductions in physical fitness are needed to reduce injury risk during the pre-season, as suggested in a study conducted over one year in youth female soccer players [15]. Indeed, the start of a new season (pre-season) implies a great initial impact on players due to the rapid increase in the volume and intensity of training [16]. Such a sudden impact of load on players can be dangerous (e.g., it can increase injury risk), especially in the cases of low fitness status (e.g., promoted by DT) [17]. It has been suggested that the maintenance of good fitness levels can moderate spikes in load (which naturally occur during the pre-season), thus decreasing the likelihood of injury [18].

Soccer performance is multidimensional [19], with physical fitness being a cornerstone supporting technical, tactical, and decision-making abilities during matches. Soccer players cover distances of up to 14 km while performing short-duration maximal-to-near-maximal actions (e.g., jumping, accelerating, sprinting, changing direction) [20, 21]. Based on the match demands, there is a need for well-prepared fitness levels, including aerobic and anaerobic capacity, as well as strength and power [22]. The determinant capacities sustaining match demands can be highlighted as maximal oxygen uptake (VO_{2max}), the capacity to repeatedly perform high-intensity exercises (commonly measured using the Yo–Yo intermittent recovery test—YYIRT), lower-limb power (commonly measured using countermovement jump), linear sprinting speed (commonly measured using sprinting time), or repeated sprint ability [23]. In addition to fitness status, body composition with small values of body fat is also recommended, considering that a greater body fat can be detrimental to a player's performance. In fact, body fat can act as "dead" weight when performing common game activities, such as running and jumping, in which one's body mass must be repeatedly lifted against gravity [24].

Considering the potentially detrimental effects of DT on physical fitness and body composition during the off-season, it seems relevant to summarize the evidence of previous randomized controlled trials and cohort studies that might help to identify the extent of DT among soccer players. A systematic review of this topic [1] highlighted the changes that occur in physical qualities during the off-season. However, this systematic review did not conduct a meta-analysis, nor did it compare the effects of training cessation vs. OTP. Thus, to systematize this knowledge, a systematic review with a meta-analysis that allows for the proper quantification of DT effects in soccer players exposed to training cessation or OTP is needed. Therefore, the purpose of the present systematic review and meta-analysis was to assess the effects of DT on men soccer players in terms of body fat, VO_{2max}, distance covered in the YYIRT, sprinting time, vertical jump height (VJH), and repeated sprint ability. Particular comparisons of DT effects between training cessation and OTPs are performed.

2 Methods

This systematic review and meta-analysis followed the Cochrane Collaboration guidelines [25]. The systematic review strategy was conducted according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines [26]. The protocol was submitted to PROSPERO with the identification number of PROSPERO 2020 CRD42020182493.

2.1 Information Sources

Electronic databases (Web of Science, Scopus, SPORTDiscus and PubMed) were searched for relevant publications prior to the 13th April 2020. Keywords and synonyms were entered in various combinations (i.e., "Soccer" OR "Football" AND Detraining OR "transition period" OR "off-season" OR "training cessation" OR "seasonal alterations").

2.2 Eligibility Criteria

The a priori inclusion criteria for this review were as follows: (1) detraining periods of at least 2 weeks; (2) detraining periods combining training cessation (of at least 2 weeks) and OTP in the following weeks; (3) detraining periods with only training cessation; (4) controlled trials or cohorts of healthy men soccer players with no restriction for age or competitive level; (5) articles written in English; (6) only full-text and original articles; and (7) only studies in off-season period.

Studies were excluded on the basis that they: (1) included other sports; (2) did not analyzed off-season; (3) presented cross-sectional (transversal) designs; (5) were review articles, letter or editorials, errata, invited commentaries or conference abstracts.

A posteriori inclusion criteria were: (1) a measure of body fat (%), VO_{2max} (ml kg^{-1} min^{-1}), YYIRT distance (meters), sprinting time (time in seconds), VJH (countermovement jump measured in centimeters), and repeated sprint ability (total time in seconds). In the specific case of YYIRT, both versions of the tests (Level 1 and Level 2) were considered and were adjusted to different competitive levels. However, for sprinting, only the longest distance reported in each study was collected.

2.3 Extraction of Data

A data extraction sheet, adapted from Cochrane Consumers and Communication Review Group's data extraction template [27], was used to assess inclusion requirements and subsequently tested on ten randomly selected studies (i.e., pilot testing). This process was conducted by two independent reviewers (HS, FMC). Any disagreement regarding study eligibility was resolved by a third reviewer (RRC). Full text articles excluded, with reasons, were recorded. The records were registered in a form created in Microsoft Excel (Microsoft Corporation, Readmon, WA, USA).

2.4 Data items

Aiming to establish consistency in data analyzing and reporting, only measures that were analyzed two or more times for different articles were included. Body fat was chosen as a measure of body composition. Body fat was expressed as a percentage and the following three methods of determination were allowed for inclusion: (1) dual-energy X-ray absorptiometry (DXA); (2) bioelectrical impedance analysis; and (3) skinfold thickness measurement. The VO_{2max} (e.g., maximal oxygen uptake; peak oxygen uptake) expressed as mL kg^{-1} min^{-1} was chosen as a measure of cardiorespiratory fitness and was included following: (1) direct determination using gas analyzer in a graded exercise test in which exercise intensity is progressively increased until exhaustion; and (2) estimation using valid field-based tests (e.g., multistage fitness test). The YYIRT distance (meters) was identified and summarized during both levels 1 and 2. Sprinting time, expressed as time (seconds), was collected from different distances while keeping the same condition of a linear trajectory and recording from the beginning until the end of the trial (no partial times were collected). Only the longest distance was collected in each study. The countermovement jump test (regular or with arms) was chosen to express the vertical height jump (cm) of players. Repeated sprint ability was expressed in total sprinting time (sec) and was derived from different protocols, although including all linear sprinting and trials interspaced by a period of recovery.

In addition to the above-mentioned data, the following information was also extracted from the studies: (1) type of detraining period (training cessation with no structure exercise; or OTP: individual training program for off-season period), length of the period (number of weeks) and moment of the season (off-season); (2) competitive level of players (youth; regional/amateur; semiprofessional; professional); and (3) characterization of the participants (age, competitive level, anthropometrics, and fitness status).

2.5 Assessment of methodological quality

The assessment of methodological quality of the included articles was conducted using the STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) checklist for cohort studies [28]. This checklist scores 22 items, in the categories of title and abstract (item 1), introduction (items 2–3), methods (items 4–12), results (item 13–17), discussion (items 18–21), and other information (item 22). Two of the authors (FMC and HM) independently screened and rated the included full articles. The agreement of both authors was tested using the k agreement rate. Cohen's kappa coefficient (k) revealed a k agreement of $k = 0.84$.

2.6 Summary measures

The analysis and interpretation of results in this meta-analysis were only conducted when at least three study groups provided baseline and follow-up data for the same measure

[29–31]. Means and standard deviations for a measure (body fat; VO_{2max}; YYIRT distance; vertical height jump; sprinting time; repeated sprint ability) of pre–post training cessation or OTP were converted to Hedges's *g* effect size (ES). The inverse variance random-effects model for meta-analyses was used because it allocates a proportionate weight to trials based on the size of their individual standard errors [32] and enables analysis while accounting for heterogeneity across studies [33]. Thus, the likelihood approach with random effects was used to better account for the inaccuracy in the estimate of between-study variance [34]. The ESs were presented alongside 95% confidence intervals (CIs). The calculated ES were interpreted using the thresholds outlined for standardized mean difference [35]: < 0.2, trivial; 0.2–0.6, small; > 0.6–1.2, moderate; > 1.2–2.0, large; > 2.0–4.0, very large; > 4.0, extremely large. In some studies in which there was more than one intervention group, the control group was proportionately divided to facilitate comparison across all participants [36]. All analyses were carried out using the Comprehensive Meta-Analysis program (version 2; Biostat, Englewood, NJ, USA). For the analysis of studies with a within-group design, when the correlations between the pre- and post-treatment measures were not available, we used a conservative estimate ($r = 0.7$), according to the previous recommendations [37, 38].

2.7 Synthesis of Results

To estimate the degree of heterogeneity between the included studies, the percentage of total variation across the studies due to heterogeneity (Cochran's *Q* statistic) [39] was used to calculate the I^2 statistic. This represents the proportion of effects that are due to heterogeneity as opposed to chance [40]. Low, moderate, and high levels of heterogeneity correspond to I^2 values of < 25%, 25–75%, and > 75%, respectively [39, 41]. Nevertheless, these thresholds are considered tentative [39]. The Chi square test assesses if any observed differences in results are compatible with chance alone. A low *p* value, or a large Chi square statistic relative to its degree of freedom, provides evidence of heterogeneity of intervention effects beyond those attributed to chance [32].

2.8 Risk of Bias Across Studies

The extended Egger's test [42] was used to assess the risk of bias across the studies. With the purpose of reducing the risk of heterogeneity, sensitivity analyses were performed aiming to determine the robustness of the summary estimates. Thus, in order to examine the effects of each result from each study on the overall findings, the results were analyzed with each study deleted from the model once. It is acknowledged that other factors, such as differences in trial quality or true study heterogeneity, could produce asymmetry.

3 Results

3.1 Study Identification and Selection

Searching of databases identified an initial 563 titles. These studies were then exported to reference manager software (EndNote™ X9, Clarivate Analytics, Philadelphia, PA, USA). Duplicates (247 references) were subsequently removed either automatically or manually. The remaining 316 articles were screened for their relevance based on titles and abstracts, resulting in the removal of a further 265 studies. The full texts of the remaining 51 articles were examined diligently; 20 were rejected as they did not satisfy the relevant criteria (detraining in men soccer players). Following the screening procedure, 31 articles were selected for in depth reading and analysis. After reading the full texts, a further 19 studies were excluded owing to a number of reasons including: in-season period ($n = 1$); conducted in other sports ($n = 3$); no clarification of methodology and detraining characteristics ($n = 2$); no inclusion of the selected measures ($n = 6$); no analysis of the detraining period in the seasonal variations ($n = 4$); were conference abstracts or just abstracts ($n = 2$); and were written in languages other than English ($n = 1$). Therefore, twelve articles were eligible for the systematic review and meta-analysis (Fig. 1). The twelve studies included provided mean and standard deviation post-detraining data for at least one main outcome.

3.2 Study Characteristics

The characteristics of the twelve studies included in the meta-analysis can be found in Table 1. In addition, the details of OTP for those studies that used this approach can be found in Table 2. Moreover, detraining periods varied between 2 and 12 weeks across the studies. Seven studies were conducted in professional players, three studies were conducted in semi-professional levels, and two in young/youth. Two of the included studies [43, 44] applied an intense intervention consisting of five high-intensity sessions per week for two weeks. Both were included as OTP. However, the characteristics of such programs are not feasible in the off-season. Nevertheless, they are included according to the logic of comparing training cessation with training interventions during the end of the season.

3.3 Methodological Quality

Table 3 presents the summary of the STROBE statement checklist. From the twelve included studies in the meta-analysis, five studies were classified between 14 and 15, five between 16 and 17 and two with 18 points.

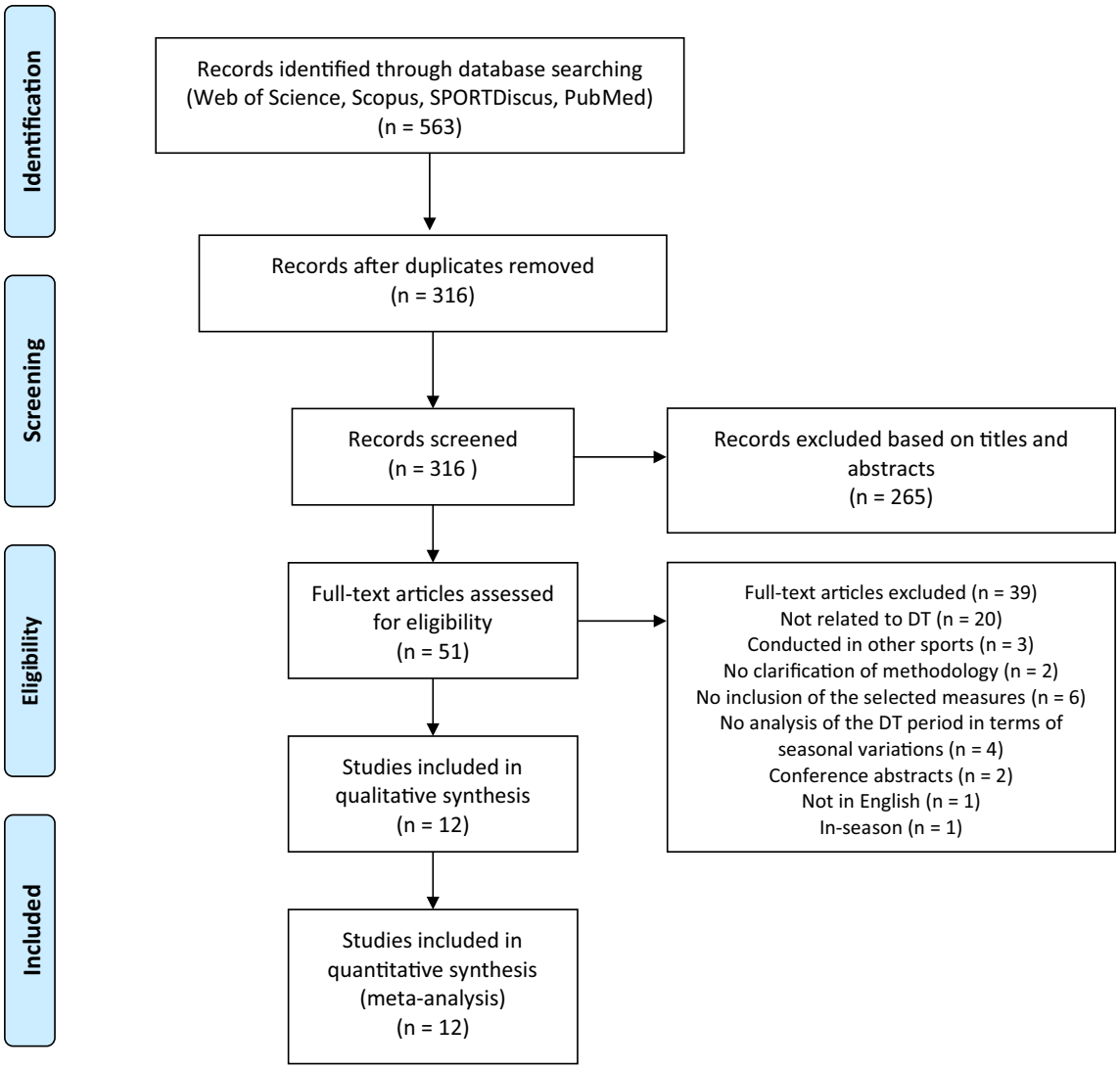

Fig. 1 PRISMA flow diagram highlighting the selection process for the studies included in the current systematic review. *DT* detraining

3.4 Effects of Training Cessation and OTP on Body Fat

Significant increases in body fat were observed after both training cessation (four study groups; ES = 0.26 [95% CI 0.07–0.46], $p = 0.009$; within group $I^2 = 0.0\%$; Egger's test = 0.718) and OTP (seven study groups; ES = 0.32 [95% CI 0.21–0.43], $p < 0.001$; within group $I^2 = 0.0\%$; Egger's test = 0.892), without differences between them (between group $p = 0.602$). The relative weight of each study group in the training cessation and OTP analyses varied between 18.4 and 32.8%, and between 5.1 and 33.6%, respectively (indicated by the size of the plotted box in Fig. 2a).

From the included studies (Table 4), two provided body fat data for both pre–post training cessation (i.e., control) and pre–post OTP (i.e., experimental). The relative weight of each study in the analysis varied between 28.6 and 71.4% (indicated by the size of the plotted box in Fig. 2b). There was no significant difference regarding body fat changes between training cessation and OTP (ES = 0.30 [95% CI −0.15 to 0.75], $p = 0.194$; within group $I^2 = 0.0\%$). Due to the limited number of studies, the heterogeneity analysis was not performed.

3.5 Effects of Training Cessation and OTP on Maximal Oxygen Uptake

A summary of the studies included in this section can be found in Table 5. Significant decreases in VO_{2max} were observed after both training cessation (three study groups; ES = −1.475 [95% CI −1.991 to −0.960], $p < 0.001$; within group $I^2 = 59.2\%$; Egger's test = 0.050, with unchanged results after adjusting according to Duval and Tweedie´s method) and OTP (four study groups; ES = −0.481 [95%

Table 1 Characteristics of the included studies

Study	Age (yo) Mean ± SD	Sample size (*n*)	Competitive level	Design	Detraining period (weeks)	Detraining characteristics	Period of the season	Measures	Tests
Caldwell et al. [48]	TC: 24 ± 4.4	TC: 13	Semiprofessional	Cohort	12 weeks (not specifically mentioned)	ND	Off-season	Body fat (%) Vertical jump (cm) VO_{2max} (mL·kg^{-1}·min^{-1}) Sprint time (s)	Bioelectrical impedance analyser Countermovement jump (arm swing) Multi-stage fitness test 15-m sprinting
Joo [51]	TC: 22.1 ± 1.8 OTP: 22.1 ± 1.8	TC: 10 OTP: 10	Semiprofessional	Cohort	2 weeks	Training cessation OTP	Off-season	Body fat (%) Sprint time (s) RSA total time (s) Distance (m)	Bioelectrical impedance analyser 30-m sprinting Repeated sprint test (7 sprints of 34.2 m interspersed by 25 s) Yo–Yo Intermittent Recovery test (L2)
Koundourakis et al. [11]	OTP: 25.6 ± 6.2	OTP: 67	Professional	Cohort	6 weeks	Training cessation + OTP	Off-season	Body fat (%) Vertical jump (cm) VO_{2max} (mL·kg^{-1}·min^{-1}) Sprint time (s)	Skinfold thickness measurement Countermovement jump Gas analyzer at a treadmill 20-m sprinting
Melchiorri et al. [47]	TC: 15 ± 1	TC: 14	Youth	Cohort	6 weeks	Training cessation	Off-season	VO_{2max} (mL·kg^{-1}·min^{-1})	Gas analyzer at a treadmill
Nakamura et al. [52]	TC: 22.0 ± 0.0 OTP[a]: 23.3 ± 2.8 OTP[b]: 22.7 ± 2.4	TC: 5 OTP[a]: 13 OTP[b]: 11	Semiprofessional and regional league	Controlled trial	3 weeks	Training cessation (control group) OTP (two groups)	Off-season	Distance (m) Sprint time (s)	Yo–Yo Intermittent Recovery test (L2) 20-m sprinting
Requena et al. [9]	OTP: 26.2 ± 2.8	OTP: 19	Professional	Cohort	6 weeks	Training cessation + OTP	Off-season	Body fat (%) Vertical jump (cm) Sprint time (s)	Bioelectrical impedance analyser Countermovement jump 30-m
Sotiropoulos et al. [14]	TC: 24.4 ± 3.0 OTP: 23.2 ± 2.6	TC: 20 OTP: 38	Professional	Controlled trial	4 weeks	Training cessation (control group) OTP	Off-season	Body fat (%) VO_{2max} (mL·kg^{-1}·min^{-1})	Skinfold thickness measurement Gas analyzer at a treadmill
Suarez-Arrones et al. [8]	OTP: 27.3 ± 2.8	OTP: 10	Professional	Cohort	5 weeks	Training cessation + OTP	Off-season	Body fat (%)	DXA
Thomassen et al. [44]	TC: 23.4 ± 0.8 OTP: 23.4 ± 0.8	TC: 11 OTP: 7	Professional	Controlled trial	2 weeks	Training cessation (control group) OTP	Off-season	Distance (m) RSA total time (s)	Yo–Yo Intermittent Recovery test (L2) Repeated sprint ability (10 sprints of 20-m sprints interspaced by 15 s)

Table 1 (continued)

Study	Age (yo) Mean ± SD	Sample size (*n*)	Competitive level	Design	Detraining period (weeks)	Detraining characteristics	Period of the season	Measures	Tests
Vassilis et al. [12]	TC: 15.1 ± 0.3	TC: 13	Young	Cohort	4 weeks	Training cessation[c]	Off-season	Body fat (%)	Skinfold thickness measurement
Koundourakis et al. [49]	OTP[d]: 25.5 ± 5.3 OTP[e]: 24.7 ± 4.9	OTP[d]: 23 OTP[e]: 22	Professional	Cohort	6 weeks	Training cessation + OTP	Off-season	Body fat (%) Vertical jump (cm) VO_{2max} ($mL \cdot kg^{-1} \cdot min^{-1}$) Sprint time (s)	Skinfold thickness measurement Countermovement jump Gas analyzer at a treadmill 20-m sprinting
Christensen et al. [43]	TC: 23.4 ± 3.5 OTP: 23.4 ± 3.5	TC: 11 OTP: 7	Professional	Controlled trial	2 weeks	Training cessation (control group) OTP	Off-season	Distance (m) RSA total time (s)	YYIRT (L2) Repeated sprint ability (10 sprints of 20-m sprints interspaced by 15 s)

ND not detailed, *TC* training cessation, *OTP* off-season training program, *HIT* high intensity interval training, *DXA* dual-energy X ray absorptiometry, *sec* seconds, *L1* level 1, *L2* level 2, *RSA* repeated sprint ability, *yo* years old, *SD* standard deviation

[a]Moderate running group

[b]Plyometric training

[c]Data from the training cessation period

[d]Team A

[e]Team B

Table 2 Characteristics of the included studies DT with off-season training programs

	Duration of training cessation (w)	Duration of OTP (w)	Frequency of OTP (d/w)	Frequency of CR (d/w)	Frequency of PT/RT (d/w)	CR: Session duration (min)	CR: Training Regimen	CR: Intensity	PT: sets/reps/exercise	RT: sets/reps/exercise	RT: intensity
Joo [51]	0	2	3	3	0	ND	3 × 12 min running	80–90% HRmax	–	–	–
Koundourakis et al. [11]	2	2	3	3	0	20-30 min	(30, 20, 2 × 15, 3 × 10, 2 × 10)	50–60% VO_{2max}	–	–	–
Nakamura et al. [52][a]	0	3	2	2	0	60 min	30 min continuous	70–80% HRmax	–	–	–
Nakamura et al. [52][b]	0	3	2	0	2	–	–	–	3/2 × 10 square jumps (60 jumps) 3/2 × 16.5 m bounding jumps (60 jumps)	–	–
Requena et al. [9]	2	4	5	3	2	ND	25–40 min continuous or 3 sets of 6–8 min HIIT	60–70% Vvam-Eval in continuous or 40–110% in HIIT	–	3–5/6–8/squat, bench press, chin-ups and dips	Body weight/ 60–70% body 1RM
Sotiropoulos [14][c]	0	4	~ 5	3	2	ND	15–20 min continuous	Jogging	-	2–4/8–20/ bench press, biceps curls, calfs, pull-ups, abdominals, lower back, quadriceps stretches	Body weight/ ND
Suarez-Arrones et al. [8]	2	3	4	4	2	ND	Long and short HIIT	90–95% MAS	–	3 × 6–8/leg press, push-up TRX, pull-TRX, single deadlift with dumbbell, standing dumbbell fly on fitball, barbell lunge split jumps	Body weight/ ND
Thomassen et al. [44]	0	2	5	5	0	ND	8 × 2 min SSGs and 10–12/25 to 30 s all out sprinting	88.4 ± 1.7% HRmax	–	–	–

Table 2 (continued)

	Duration of training cessation (w)	Duration of OTP (w)	Frequency of OTP (d/w)	Frequency of CR (d/w)	Frequency of PT/RT (d/w)	CR: Session duration (min)	CR: Training Regimen	CR: Intensity	PT: sets/reps/exercise	RT: sets/reps/exercise	RT: intensity
Koundourakis et al. [49]	2	2	3	0	0	20-30 min	(30, 20, 2×15, 3×10, 2×10)	50–60% VO_{2max}	–	–	–
Christensen et al. [43][d]	0	2	5	5	0	ND	8×2 min SSGs and 10–12/25 to 30 s all out sprinting	88.4 ± 1.7% HRmax	-	-	-

OTP off-season training program, *w* number of weeks, *d* number of days per week, *CR* cardiorespiratory training while running, *HRmax* maximum heart rate, VO_{2max} maximal oxygen uptake, *PT* plyometric training, *RT* resistance training, *m* meters, *ND* not defined, *HIIT* high-intensity interval training, *1RM* one-repetition maximum, *sec* seconds, *min* minutes, *MAS* maximal aerobic speed, *SSGs* small-sided games

[a]Moderate running group

[b]Plyometric training group

[c]Combined off-season program

[d]HITG: high-intensity training group

CI −0.821 to −0.142], $p = 0.005$; within group $I^2 = 83.0\%$; Egger's test = 0.371). However, the decrease was greater after training cessation as compared to OTP (between group $p = 0.002$). The relative weight of each study group in the training cessation and OTP analyses varied from 21.9 to 28.3%, and from 26.3 to 38.6%, respectively (indicated by the size of the plotted box in Fig. 3).

3.6 Effects of Training Cessation and OTP on Yo–Yo Intermittent Recovery Test Distance

A significant decrease in YYIRT distance was observed after training cessation (4 study groups; ES = −0.461 [95% CI −0.755 to −0.168], $p = 0.002$; within group $I^2 = 50.2\%$; Egger's test = 0.605), although not after OTP (5 study groups; ES = −0.062 [95% CI −0.665 to 0.542], $p = 0.842$; within group $I^2 = 86.4\%$; Egger's test = 0.616). The changes were not significantly different between training cessation compared to OTP (between group $p = 0.242$). The relative weight of each study group in the training cessation and OTP analyses varied from 17.6 to 34.0% and from 19.1 to 20.8%, respectively (indicated by the size of the plotted box in Fig. 4a).

From the included studies (Table 6), 4 provided YYIRT distance data for both pre–post training cessation (i.e., 4 control groups) and pre–post OTP (i.e., 5 experimental groups). The relative weight of each study in the analysis varied between 18.3 and 21.8% (indicated by the size of the plotted box in Fig. 4b). There was no significant difference regarding YYIRT changes between training cessation and OTP (ES = 0.553 [95% CI −0.224 to 1.331], $p = 0.163$; within group $I^2 = 75.0\%$; Egger's test = 0.009 was unchanged after adjusting according to Duval and Tweedie's method).

3.7 Effects of Training Cessation and OTP on Vertical Height Jump

Significant decreases in vertical jump height were observed after both training cessation (one study group; ES = −0.806 [95% CI −1.290 to −0.321], $p = 0.001$; within group $I^2 = 0.0\%$) and OTP (four study groups; ES = −0.508 [95% CI −0.716 to −0.300], $p < 0.001$; within group $I^2 = 46.3\%$; Egger's test = 0.968), without significant differences between them ($p = 0.268$). The relative weight of each study group in the OTP analyses varied between 19.8 and 36.5 (indicated by the size of the plotted box in Fig. 5). For training cessation analyses, the relative weight was 100% (one study only). A summary of the studies can be found in Table 7.

3.8 Effects of Training Cessation and OTP on Sprinting Time

A nonsignificant increase in sprint time was observed after training cessation (3 study groups; ES = 0.524 [95%

Table 3 STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) checklist for cohort studies

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	Total
Caldwell et al. [48]	0	1	1	1	1	1	1	1	1	0	1	1	0	0	1	0	0	1	1	1	1	0	15
Joo [51]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	0	0	18
Koundourakis et al. [11]	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	1	0	0	16
Melchiorri et al. [47]	0	1	1	1	1	1	1	1	1	0	1	1	1	0	1	0	0	1	0	1	0	0	14
Nakamura et al. [52]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	0	1	0	0	15
Requena et al. [9]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	0	1	0	0	15
Sotiropoulos et al. [14]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	0	1	0	0	15
Suarez-Arrones et al. [8]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1	1	0	1	18
Thomassen et al. [44]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	0	1	0	1	16
Vassilis et al. [12]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	0	16
Koundourakis et al. [49]	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	0	0	17
Christensen et al. [43]	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	0	1	0	1	16

1: Title and abstract; 2: background/rationale; 3: objectives; 4: study design; 5: setting; 6: participants; 7: variables; 8: data sources/measurement; 9: bias; 10: study size; 11: quantitative variables; 12: statistical methods; 13: participants; 14: descriptive data; 15: outcome data; 16: main results; 17: other analyses; 18: key results; 19: limitations; 20: interpretation; 21: generalizability; 22: funding

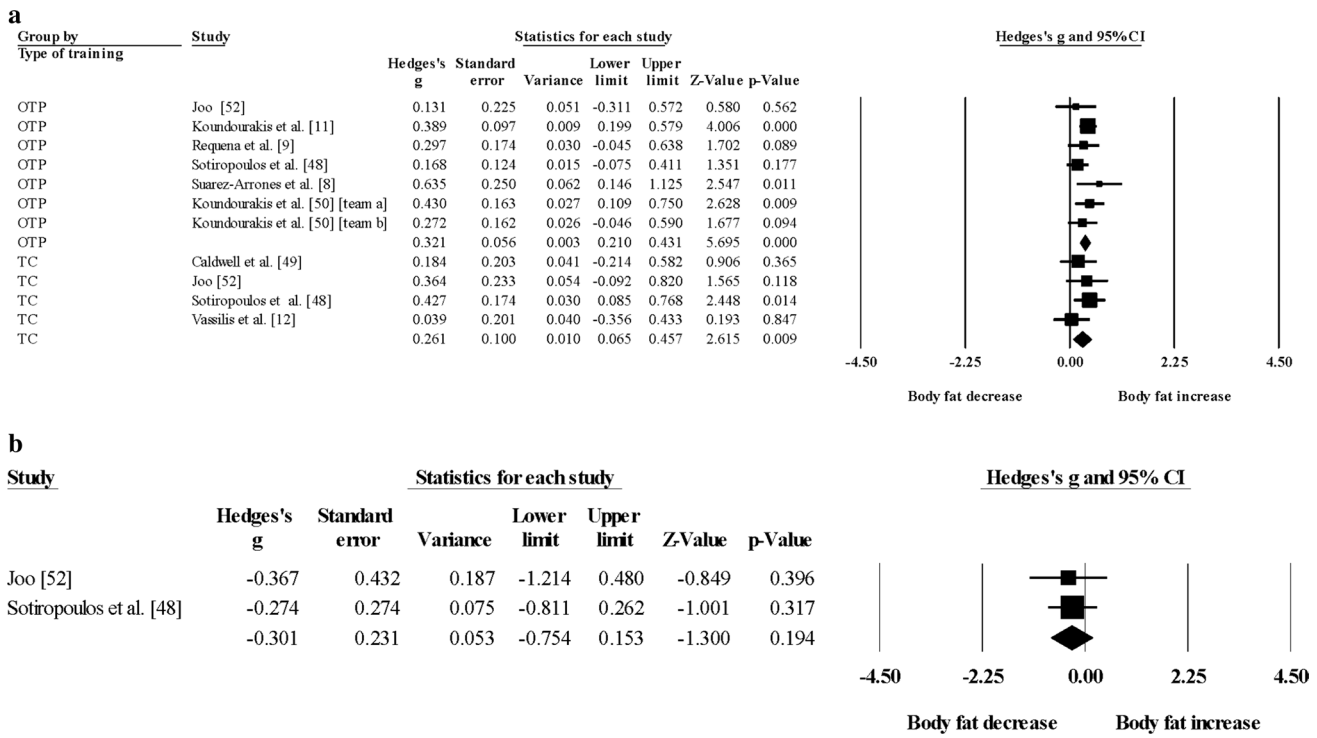

Fig. 2 **a** Forest plot of changes in body fat (%) in male soccer players after training cessation (TC) or after an off-season training program (OTP). Values shown are effect sizes with 95% confidence intervals (CI). Black boxes: individual study groups. Black diamonds: subgroup overall. **b** Same as (**a**) for the two studies that included both a training cessation period and an OTP (negative values reflect the comparative between groups, not a reduction in body fat percentage). Black diamond: overall result. Note: the relative weight of each study in the analyses is indicated by the size of the plotted box in each figure

Table 4 Summary of the included studies and results of body fat (%) reported before and after detraining period

Study	Before Mean ± SD	After Mean ± SD	Before–after (Δ%)
Training cessation			
Caldwell et al. [48]	14.3 ± 3.1	14.9 ± 3.0	4.2
Joo [51][a]	9.3 ± 1.2	9.8 ± 1.3	5.4
Sotiropoulos et al. [14][a]	7.8 ± 1.8	8.6 ± 1.8	10.3
Vassilis et al. [12][b]	14.6 ± 2.3	14.7 ± 2.5	0.7
OTP			
Joo [51][c]	9.6 ± 0.7	9.7 ± 0.7	1.0
Koundourakis et al. [11][d]	8.8 ± 3.0	10.05 ± 3.5	14.8
Requena et al. [9][d]	10.5 ± 3.5	11.6 ± 3.6	10.5
Sotiropoulos et al. [14][d]	7.9 ± 1.7	8.2 ± 1.8	3.8
Suarez-Arrones et al. [8][c]	14.6 ± 0.9	15.5 ± 1.4	6.2
Koundourakis et al. [49][d,e]	9.3 ± 3.3	11.0 ± 4.1	18.3
Koundourakis et al. [49][d,f]	9.3 ± 3.6	10.4 ± 4.1	11.8

OTP off-season training program, *DT* detraining, *SD* standard deviation, Δ% percentage of difference

[a]Control group

[b]Data from the training cessation period

[c]High-intensity exercise

[d]Combined off-season program

[e]Team A

[f]Team B

Table 5 Summary of the included studies and results of VO_{2max} ($mL \cdot kg^{-1} \cdot min^{-1}$) reported before and after detraining period

Study	Before DT Mean ± SD	After DT Mean ± SD	Before–after DT (Δ%)
Training cessation			
Caldwell et al. [48]	58 ± 1.9	56 ± 1.2	−3.4
Melchiorri et al. [47]	62.8 ± 5.8	49.5 ± 6.5	−21.2
Sotiropoulos et al. [14][a]	58.1 ± 2.6	54.5 ± 2.8	−6.2
OTP			
Koundourakis et al. [11][b]	59.4 ± 3.1	58.9 ± 3.5	−0.8
Sotiropoulos et al. [14][b]	57.7 ± 2.6	56.9 ± 2.5	−1.4
Koundourakis et al. [49][b,c]	60.3 ± 2.5	57.7 ± 2.5	−4.3
Koundourakis et al. [49][b,d]	60.5 ± 4.1	58.3 ± 3.9	−3.6

OTP off-season training program, *DT* detraining, *SD* standard deviation, Δ% percentage of difference

[a]Control group

[b]Combined off-season program

[c]Team A

[d]Team B

CI −0.094 to 1.142], $p = 0.097$; within group $I^2 = 85.0\%$; Egger's test = 0.075), while a significant increase was observed after OTP (seven study groups; ES = 0.862 [95% CI 0.702–1.022], $p < 0.001$; within group $I^2 = 21.3\%$; Egger's test = 0.214). No significant differences were observed between training cessation and OTP (between group $p = 0.299$). The relative weight of each study group in the training cessation and OTP analyses varied from 30.5 to 36.8% and from 8.4 to 29.0%, respectively (indicated by the size of the plotted box in Fig. 6a).

Two studies (involving 3 experimental groups) provided sprint time data for both pre–post training cessation (i.e., control) and pre–post OTP (i.e., experimental) (Table 8). The relative weight of each study in the analysis varied between 29.0 and 37.5% (indicated by the size of the plotted box in Fig. 6b). There was no significant difference regarding sprint time changes between training cessation and OTP (ES = 0.423 [95% CI −0.029 to 0.874], $p = 0.067$; within group $I^2 = 0.0\%$; Egger's test = 0.279).

3.9 Effects of Training Cessation and OTP on Repeated Sprint Ability

Significant increase in repeated sprint ability total time was observed after training cessation (3 study groups; ES = −0.688 [95% CI −0.984 to −0.394], $p < 0.001$; within group $I^2 = 0.0\%$; Egger's test = 0.816), although not after OTP (3 study groups; ES = 0.274 [95% CI −0.081 to 0.628], $p = 0.130$; within group $I^2 = 33.6\%$; Egger's test = 0.005, with unchanged results after adjusting with Duval and Tweedie's method). The changes were significantly different between training cessation as compared to OTP (between group $p < 0.001$). The relative weight of each study group in the training cessation and OTP analyses varied from 29.4 to 40.2% and from 30.3 to 39.2%, respectively (indicated by the size of the plotted box in Fig. 7a).

From the included studies (Table 9), 3 provided RAS time data for both pre–post training cessation (i.e., 3 control groups) and pre–post OTP (i.e., 3 experimental groups). The relative weight of each study in the analysis varied between 29.3 and 36.6% (indicated by the size of the plotted box in Fig. 7b). There was a significant difference regarding repeated sprint ability change between training cessation and OTP in favour of the latter (ES = −1.818 [95% CI −3.276 to −0.361], $p = 0.014$; within group $I^2 = 80.0\%$; Egger's test = 0.159).

4 Discussion

This systematic review with a meta-analysis aimed to assess the effects of training cessation in off-season training programs on male soccer players' body fat, VO_{2max}, YYIRT,

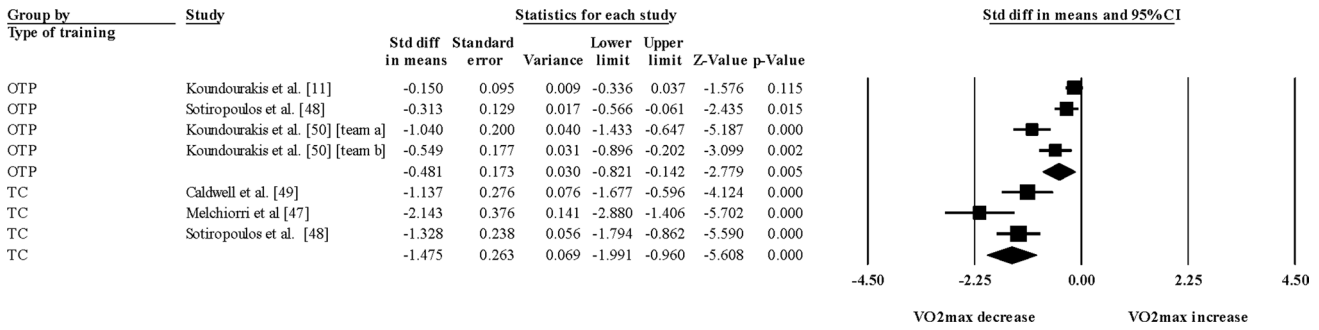

Fig. 3 Forest plot of changes in maximal oxygen consumption (VO_{2max}) in male soccer players after training cessation (TC) or after an off-season training program (OTP). Values shown are effect sizes with 95% confidence intervals (CI). Black boxes: individual study groups. Black diamonds: subgroup overall. Note: the relative weight of each study in the analyses is indicated by the size of the plotted box

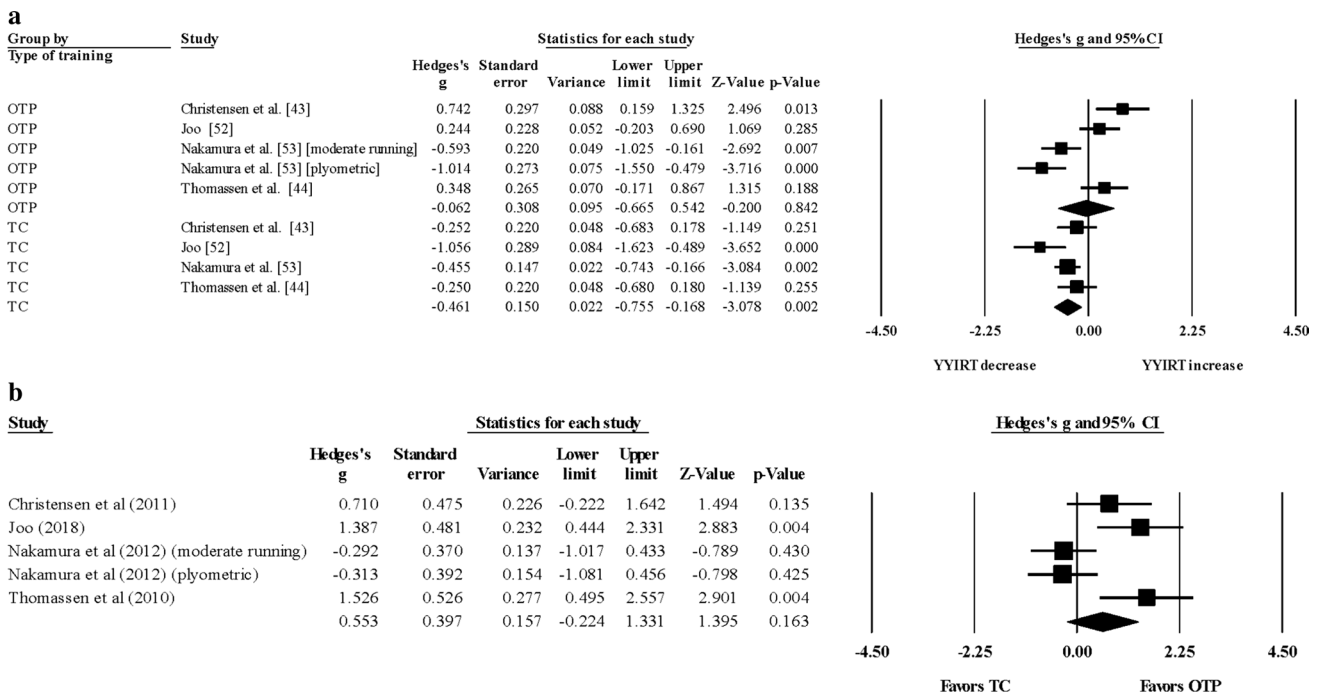

Fig. 4 **a** Forest plot of changes in the yo–yo intermittent recovery test (YYIRT) distance in male soccer players after training cessation (TC) or after an off-season training program (OTP). Black boxes: individual study groups. Black diamonds: subgroup overall. **b** Forest plot of changes in YYIRT distance in male soccer players after a TC period compared to those that completed an OTP. Black diamond: overall result. Notes for both figures: the relative weight of each study in the analyses is indicated by the size of the plotted box in each figure. Values shown are effect sizes with 95% confidence intervals (CI)

vertical jump, sprinting time, and repeated-sprint ability. In short, body fat, VO_{2max}, YYIRT, vertical jump, and repeated sprint ability were significantly and negatively affected by training cessation. However, even players who experienced off-season training programs were significantly and negatively affected in terms of their body fat, vertical jump, and sprinting time. The between-group analysis revealed significant detrimental effects on VO_{2max} and repeated sprint ability in participants in the training cessation condition in comparison to those subjected to off-season training programs. The discussion will be structured by the outcome.

4.1 Effects of Training Cessation and OTPs on Body Fat

Changes in body composition (namely, decreased lean mass and possibly increased body fat percentage) are expected among athletes after training cessation or a DT period [45].

Table 6 Summary of the included studies and results of aerobic capacity found in YYIRT (m) reported before and after detraining period

Study	Before DT Mean ± SD	After DT Mean ± SD	Before–after DT (Δ%)
Training cessation			
Joo [51][a]	880 ± 100	750 ± 120	− 14.8
Nakamura et al. [52][a]	720 ± 204	632 ± 151	− 12.2
Thomassen et al. [44][a]	845 ± 159	645 ± 30	− 22.6
Christensen et al. [43][a]	845 ± 160	654 ± 99	− 22.6
OTP			
Joo [51][b]	880 ± 120	910 ± 100	3.4
Nakamura et al. [52][c]	855 ± 222	724 ± 133	− 15.3
Nakamura et al. [52][d]	860 ± 120	727 ± 122	− 15.5
Thomassen et al. [44][e]	937 ± 148	994 ± 72	6.1
Christensen et al. [43][e]	937 ± 56	994 ± 72	6.1

OTP off-season training program, *DT* detraining, *SD* standard deviation, Δ%: percentage of difference

[a]Control group

[b]High-intensity exercise

[c]Moderate running group

[d]Plyometric training group

[e]HITG: high-intensity training group

Both factors (decrease in lean mass and increase in body fat percentage) may negatively affect tolerance to high training volumes and intensities during the restart of the season, thus potentially constituting a higher risk factor for muscle strain in soccer [1]. Moreover, a greater body fat percentage may also be detrimental to players' performance, considering that it can act as "dead" weight during common game activities, such as running and jumping, in which the body mass is repeatedly lifted against gravity [24].

All of the studies included in this systematic review and meta-analysis showed increases in body fat after training cessation and OTPs (Fig. 2). Overall, a small and significant increase in body fat (ES = 0.306; $p < 0.001$) was observed after training cessation and OTPs. These findings revealed that even in cases of OTPs, there is a tendency for detrimental effects on body composition to occur, suggesting that prescribing training sessions during DT periods are not enough for players to maintain their fitness status. In fact, changes in body composition during a DT period could be expected because variations in percentage body fat can be caused by a decrease in lean-mass (and, therefore, an increase in the weight of body fat in relation to body mass) [45] and an increase in body fat mass [8]. Lifestyle changes during off-season periods (e.g., nutritional behaviors and physical activity) might explain the increases in body fat during the off-season, even in players adhering to training programs [46].

4.2 Effects of Training Cessation and OTPs on Maximal Oxygen Uptake

The systematic review and meta-analysis revealed a significant decrease in VO_{2max} after training cessation and OTPs ($p < 0.05$). Among the training cessation groups [14, 47, 48], the magnitudes of change were moderate, and the percentages of declines ranged from − 3.4% in semi-professionals [48] up to − 21.2% in youth [47]. However, the results of this study in youth players (− 21.2%) [47] might inflate the final values of all studies. Thus, the magnitude of decreases in TC should be considered with caution, as should comparisons with OTP. In the case of OTP groups, declines varied from − 0.8% [14] to − 4.3% in professionals [49]. The OTP comprised 3–5 training days a week (see Table 2), all involving jogging at 50–60% VO_{2max} for 20–30 min. Possibly, the low level of stimulus imposed during the OTP was not enough to prevent significant decreases in VO_{2max}. This matter should be discussed and re-organized for the implementation of OTPs. Moderate running does not seem to be favorable enough to achieve benefits in cardiorespiratory capacity [50], and so alternative strategies should be

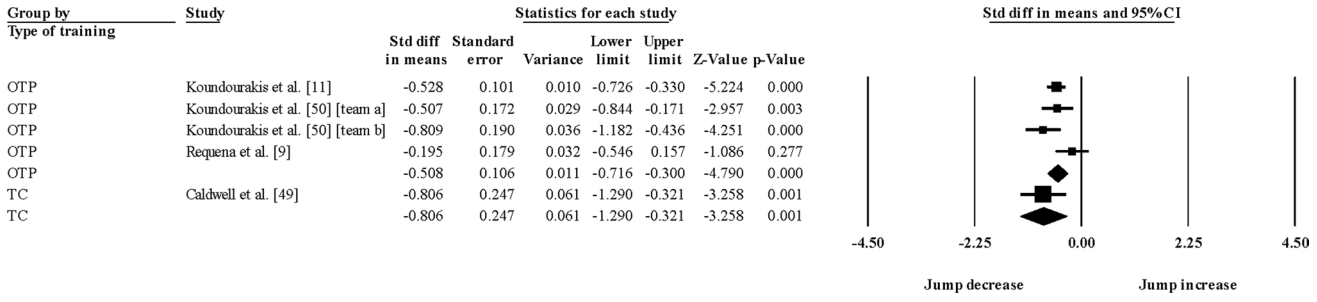

Fig. 5 Forest plot of changes in vertical jump height in male soccer players after training cessation (TC) or after an off-season training program (OTP). Values shown are effect sizes with 95% confidence intervals (CI). Black boxes: individual study groups. Black diamonds: subgroup overall. Black diamond: overall result. Note: the relative weight of each study in the analyses is indicated by the size of the plotted box

Table 7 Summary of the included studies and results of vertical height jump at countermovement jump (cm) reported before and after detraining period

Study	Before DT Mean ± SD	After DT Mean ± SD	Before–after DT (Δ%)
Training cessation			
Caldwell et al. [48]	57 ± 4.0	54 ± 3.2	− 5.3
OTP			
Koundourakis et al. [11][a]	40.9 ± 4.6	38.6 ± 4.0	− 5.6
Requena et al. [9][a]	51.5 ± 5.9	50.4 ± 5.3	− 2.1
Koundourakis et al. [49][a,b]	41.0 ± 4.0	39.1 ± 3.3	− 4.6
Koundourakis et al. [49][a,c]	42.8 ± 3.6	40.1 ± 2.8	− 6.3

OTP off-season training program, *DT* detraining, *SD* standard deviation, Δ%: percentage of difference

[a]Combined off-season program

[b]Team A

[c]Team B

employed. Further research (namely, randomized controlled trials) should be conducted to allow comparisons of the effects of training cessation (control), micro-dose training, and regular-dose training in large samples over short and long periods. It might be worth identifying how much training during the off-season is too much and determining recommended thresholds that should not be crossed. The consequences of training cessation and OTP should be tracked not only during the pre-season but also during the season. It should be analyzed whether training cessation affects players' recovery without crossing thresholds. Another related avenue of research could define the minimal effective dose of training for off-season periods.

4.3 Effects of Training Cessation and OTPs on Yo–Yo Intermittent Recovery Test Distance

The systematic review and meta-analysis revealed significant decreases in YYIRT (Level 1 and 2) distance after training cessation (− 5.2 to − 14.8%) but not after OTPs (Table 6; Fig. 4). The high-intensity nature of the OTP analyzed in this systematic review and meta-analysis [43, 44, 51] may have helped to avoid a detrimental effect on YYIRT performance. In fact, in several OTPs [43, 44, 51], meaningful improvements (3.4–6.1%) in YYIRT distance were observed. Such OTPs applied three to five sessions

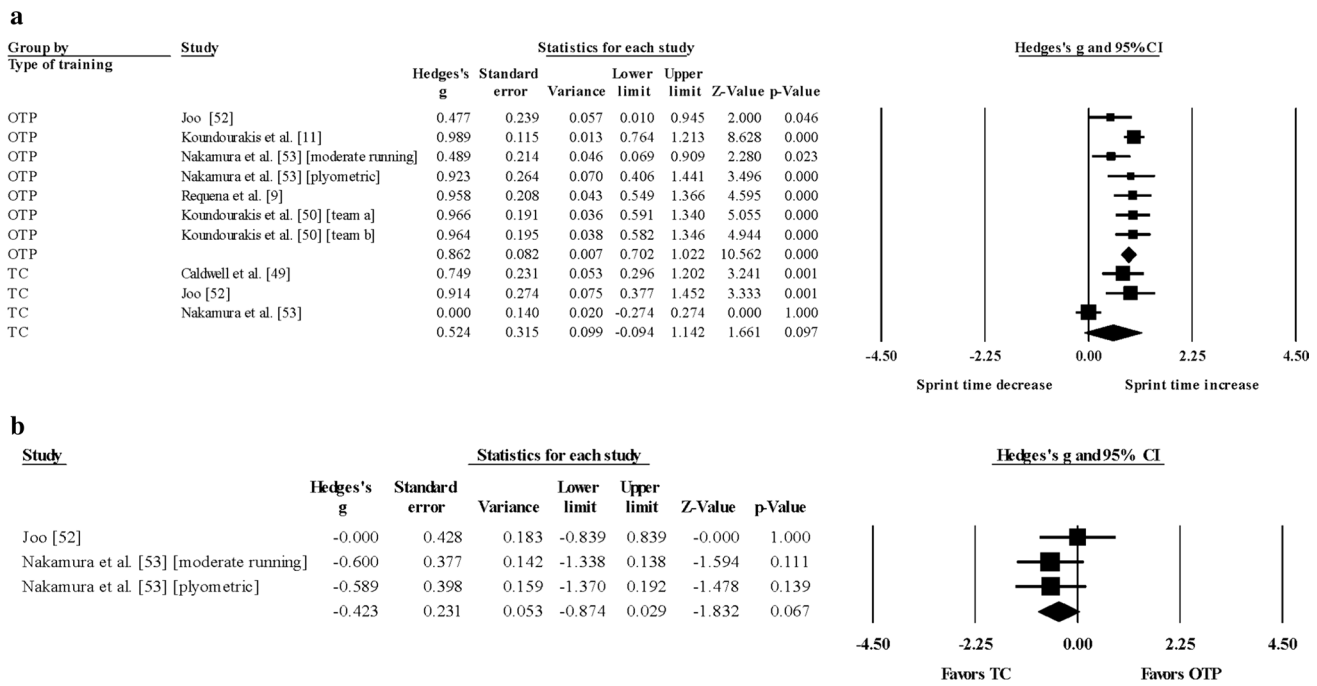

Fig. 6 **a** Forest plot of changes in sprint time in male soccer players after training cessation (TC) or after an off-season training program (OTP). Values shown are effect sizes with 95% confidence intervals (CI). Black boxes: individual study groups. Black diamonds: subgroup overall. **b** Same as (**a**) for the two studies that included both a TC period and an OTP (negative values reflect the comparative between groups, not a reduction in sprint times per se). Black diamond: overall result. Note: the relative weight of each study in the analyses is indicated by the size of the plotted box in each figure

Table 8 Summary of the included studies and results of linear sprinting (s) reported before and after detraining period

Study	Before DT Mean ± SD	After DT Mean ± SD	Before–after DT (Δ%)
Training cessation			
Caldwell et al. [48]	2.43 ± 0.09	2.51 ± 0.10	3.3
Joo [51][a]	4.2 ± 0.1	4.3 ± 0.1	2.4
Nakamura et al. [52][a]	3.1 ± 0.1	3.1 ± 0.2	0.0
OTP			
Joo [51][b]	4.1 ± 0.1	4.2 ± 0.2	2.4
Koundourakis et al. [11][c]	3.0 ± 0.1	3.1 ± 0.1	3.3
Nakamura et al. [52][d]	3.0 ± 0.2	3.1 ± 0.1	3.3
Nakamura et al. [52][e]	3.0 ± 0.1	3.1 ± 0.1	3.3
Requena et al. [9][c]	4.2 ± 0.1	4.3 ± 0.1	2.4
Koundourakis et al. [49][c,f]	3.0 ± 0.1	3.1 ± 0.1	3.3
Koundourakis et al. [49][c,g]	3.0 ± 0.1	3.1 ± 0.1	3.3

DT detraining, *SD* standard deviation, Δ% percentage of difference, *OTP* off-season training program

[a]Control group

[b]High-intensity exercise

[c]Combined off-season program

[d]Moderate running group

[e]Plyometric training group

[f]Team A

[g]Team B

per week, with regimens of 3 × 12 min (80–90%HRmax) [51] and 8 × 2 min small-sided games/10–12/25–30 s all-out sprinting (~ 88% HRmax) [43, 44]. These conditions were ensured for 2 weeks of training [43, 44, 51]. Clearly, these regimens (five high-intensity sessions in a week) and training scenarios are hard to implement under typical off-season conditions [43, 44]. In fact, these conditions can be considered as a shock-microcycle intervention and not a regular OTPs that players will adhere to. However, alternative approaches, such as micro-doses (minimal doses that can be effective), should be tested to mitigate drops in performance while also allowing players to recover from the previous season. A reduction/avoidance of the detrimental effects of DT on YYIRT performance may reduce the risk of injury in soccer players upon being reintroduced to high volumes of load during the pre-season [1]. However, it is important to consider healthy thresholds. High-intensity running has a beneficial impact on YYIRT during an OTP [43, 44, 51] while moderate running does not [52]. As such, the former (using short or long intervals) may be implemented during OTPs at least three times a week at intensities of ≥ 80% HRmax, starting from the early part of the off-season. The protocols in this work were tested only for their effects on players' capacities, not for their effects on follow-up assessments. Again, future research is required to determine how much is too much.

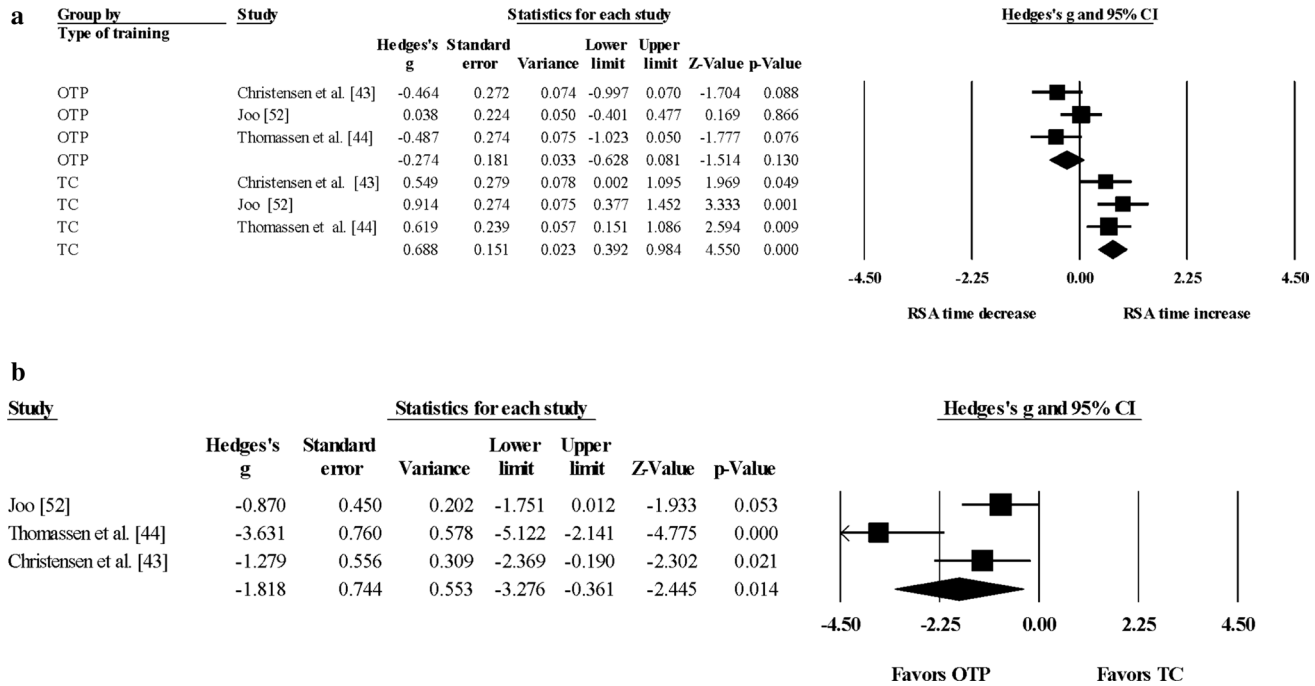

Fig. 7 **a** Forest plot of changes in repeated sprint ability (RSA) total time in male soccer players after training cessation (TC) or after an off-season training program (OTP). Black boxes: individual study groups. Black diamonds: subgroup overall, and **b** forest plot of changes in RSA total time in male soccer players after a TC period compared to those that completed an OTP. Black diamond: overall result. Notes for both figures: the relative weight of each study in the analyses is indicated by the size of the plotted box in each figure. Values shown are effect sizes with 95% confidence intervals (CI)

Table 9 Summary of the included studies and results of total time at repeated sprint ability (s) reported before and after detraining period

Study	Before DT Mean ± SD	After DT Mean ± SD	Before–after DT (Δ%)
Training cessation			
Joo [51][a]	45.9 ± 2.1	48.0 ± 2.1	4.6
Thomassen et al. [44][a]	33.4 ± 1.0	34.1 ± 0.3	2.1
Christensen et al. [43][a]	33.4 ± 1.0	34.1 ± 0.9	2.1
OTP			
Joo [51][b]	45.8 ± 2.5	45.9 ± 2.3	0.2
Thomassen et al. [44][c]	33.4 ± 1.1	32.8 ± 0.4	− 1.8
Christensen et al. [43][c]	33.4 ± 1.2	32.8 ± 1.0	− 1.8

DT detraining, *SD* standard deviation, Δ% percentage of difference, *OTP* off-season training program

[a]Control group

[b]High-intensity exercise

[c]ITG: high-intensity training group

4.4 Effects of Training Cessation and OTPs on Vertical Jump Height

The results of this systematic review and meta-analysis showed significant decreases ($p < 0.05$) in VJH (measured using countermovement jump) after OTPs and training cessation. No significant differences were found between training cessation and OTPs. The OTPs analyzed in this systematic review and meta-analysis included moderate continuous running [11, 49] or a combination of running-based exercises and slow-speed moderate-intensity resistance training [9]. Therefore, the lack of lower-limb power stimulus may help to explain the detrimental effect of OTPs on VJH. In this sense, OTPs should incorporate adequate neuromuscular stimuli (e.g., maximal strength, power–plyometric training) to reduce the detrimental effect of DT on VJH [53–55].

4.5 Effects of Training Cessation and OTPs on Sprinting Time

The results of this systematic review and meta-analysis revealed detrimental effects of training cessation and OTPs on sprinting time. No significant differences were found between training cessation and OTPs (Fig. 6). Considering that sprinting requires appropriate stimuli based on sprint-training [56], OTPs could be improved by the inclusion of specific strategies [57, 58]: (1) sprinting training at specific distances, (2) sprinting closer to maximal velocity (> 90%), and (3) determining the effects of combined interventions such as strength/power workouts.

4.6 Effects of Training Cessation and OTPs on Repeated Sprint Ability

Considering that repeated sprint ability is a complex quality that is possibly related to neuromuscular and metabolic capacity [59, 60], it is expected that DT periods will lead to a decrease in repeated sprint ability. The results of this systematic review and meta-analysis revealed a significant increase in repeated sprint ability (total time) after training cessation but not after OTP. Among the training cessation studies (Fig. 7), all showed increased repeated sprint ability total times (between 2.1 and 4.6%), while OTP studies showed either a slight increase (0.2%) [51] or an improvement (− 1.8% in two studies) [43, 44]. Repeated sprint training and high-intensity interval training during DT periods may be beneficial not only by reducing the negative effects of repeated sprint ability, but also by maintaining the capacities that sustain repeated sprint ability (e.g., oxidative capacity) [61]. However, it is important to highlight that the protocols used by two studies [43, 44] are very difficult to implement in the off-season period due to the need for recovery. In fact, they may be understood as a shock microcycle to be used in specific scenarios.

4.7 Limitations and Future Research

There are some limitations of the current systematic review and meta-analysis. The limited number of studies on some topics (e.g., VO_{2max} and VJH) did not allow us to conduct a comparison between training cessation and OTP for these factors. In addition, even for topics for which a comparison between training cessation and OTP could be performed, the results must be interpreted with caution and must be confirmed by further research.

Another limitation of this study is related to the within-group meta-analyses. Since some studies did not provide information regarding the correlation between pre–post intervention measures, we used a conservative estimate ($r = 0.7$), as previously recommended [37, 38]. In such a scenario, a focus on between-group analyses may be preferred, particularly when studies with different durations and/or participant characteristics are analyzed. However, considering the limited number of studies available, a more comprehensive overview of the reviewed topic was provided by considering within-group analyses. This was done because the between-group and within-group analyses revealed results in the same direction. Owing to the limited number of studies, it was impossible to conduct subgroup analyses for moderators, such as age group, competitive level, and duration of detraining. All these moderators are surely important, and therefore, the interpretation of the current results should

be carefully performed. However, this fact highlights the need for further research that may help us to understand the effects of detraining in different competitive scenarios and among players of different age groups and sexes.

As shown in Tables 1 and 8 out of 12 included studies were cohort studies. This represents a limitation of the current research regarding the detrimental effects of the off-season on soccer players. The absence of a comparison between groups (randomized controlled trials) or even more focused analyses of the adaptations occurring based on players' profiles substantially limits the overall quality of the methodological approach. Future research must perform more randomized controlled trials (intervention vs. control) and randomized parallel studies to test different interventions (small doses vs. low doses vs. control). In addition, confounding factors (e.g., physical activity patterns during the off-season period, dietary factors, and tracking the responders' and non-responders' profiles) need to be controlled for in future research.

Finally, it must be understood how off-season training interventions affect players across the season. Is it beneficial to return to a proper fitness status in the pre-season? How much training is too much? How does off-season training influence players' performance across the season? All these research questions must be answered in future research.

4.8 Practical Implications

The current systematic review and meta-analysis indicates some of the main detrimental effects of the off-season on training cessation and OTPs. Owing to the heterogeneity and study design of the included articles, it is hard to define an ideal strategy for OTPs. Studies that avoided decrements or even facilitated improvements included short interventions with substantial session frequencies and high-intensity methods. The issue is that no follow ups were done by the researchers to determine whether such an approach had positive or negative consequences in the medium- to long-term. Is it important to recover during the off-season? Is it important to train during the off-season? How much training is too much? The current research could not answer these questions. However, OTPs can be used to avoid high heterogeneity levels between players and to standardize fitness levels at the beginning of a new season.

Ultimately, weight-room training (e.g., strength, power, and reactive training), which is not part of in-season training, can be the core part of off-season training regimens. In this way, lower-limb power, sprinting performance, and repeated sprint ability can be sustained during the season. Additionally, some forms of cardiorespiratory training can be implemented to avoid severe drops in capacity (eventually, HIIT training once a week). However, future research should determine the minimal effective dose. Finally, instruction programs aimed to promote healthy lifestyles [62] and dietary behaviors can be introduced to ensure the maintenance of lean mass and the avoidance of increases in fat mass.

5 Conclusions

Detrimental effects on body composition and physical fitness occur after training cessation and OTPs. However, the present study indicates that OTPs seem to offer some value in alleviating the detrimental effects on VO_{2max} and repeated sprint ability. The results call for the implementation of more effective OTPs among male soccer players.

In light of the limitations of the present research, future studies should include follow-up analyses and randomized controlled trials to determine the minimal effective doses of OTP that should be applied during the off-season. Future research should also determine the medium- to long-term in-season consequences of off-season training interventions. Possibly, OTPs should be perceived as opportunities to reduce heterogeneity in the fitness levels of players and for improving physical qualities that cannot be focused on during the season (e.g., strength, power, and reactive training).

Data Availability Dataset underpinning the analysis is provided as Electronic Supplementary Material Appendix S1.

Compliance with Ethical Standards

Funding Filipe Manuel Clemente was funded by Fundação para a Ciência e Tecnologia/Miniestrério/Ministério da Ciência, Tecnologia e Ensino Superior through national funds and when applicable co-funded EU funds under the project UIDB/50008/2020. Hugo Sarmento gratefully acknowledges the support of a Spanish government subproject 'Integration ways between qualitative and quantitative data, multiple case development, and synthesis review as main axis for an innovative future in physical activity and sports research' [PGC2018-098742-B-C31] (Ministerio de Economía y Competitividad, Programa Estatal de Generación de Conocimiento y Fortalecimiento Científico y Tecnológico del Sistema I+D+i), which is part of the coordinated project 'New approach of research in physical activity and sport from mixed methods perspective' (NARPAS_MM) [SPGC201800X098742CV0]. No other specific sources of funding were used to assist in the preparation of this article.

Conflicts of interest/Competing interests Filipe Manuel Clemente, Rodrigo Ramirez-Campillo and Hugo Sarmento declare that they have no conflicts of interest relevant to the content of this review.

Acknowledgements None.

Authorship Contributions FMC led the project, ran the data search and methodological assessment, and wrote and revised the original manuscript. RRC analyzed and interpreted the data, wrote the statistical report and revised the original manuscript. HS ran the data search and methodological assessment and wrote and revised the original manuscript. All authors read and approved the final manuscript.

References

1. Silva JR, Brito J, Akenhead R, Nassis GP. The transition period in soccer: a window of opportunity. Sport Med. 2016;46:305–13. https://doi.org/10.1007/s40279-015-0419-3.
2. Jeong T-S, Reilly T, Morton J, Bae S-W, Drust B. Quantification of the physiological loading of one week of "pre-season" and one week of "in-season" training in professional soccer players. J Sports Sci. 2011;29:1161–6. https://doi.org/10.1080/02640414.2011.583671.
3. Clemente FM, Silva R, Castillo D, Arcos AL, Mendes B. Weekly load variations of distance-based variables in professional soccer players: a full-season study. Int J Environ Res Public Health. 2020;17:3300. https://doi.org/10.3390/ijerph17093300.
4. Clemente FM, Nikolaidis PT, Rosemann T, Knechtle B. Dose-response relationship between external load variables, body composition, and fitness variables in professional soccer players. Front Physiol. 2019;10:443. https://doi.org/10.3389/fphys.2019.00443/full.
5. Clemente FM, Clark C, Castillo D, Sarmento H, Nikolaidis PT, Rosemann T, et al. Variations of training load, monotony, and strain and dose-response relationships with maximal aerobic speed, maximal oxygen uptake, and isokinetic strength in professional soccer players. PLoS ONE. 2019;14:e0225522. https://doi.org/10.1371/journal.pone.0225522.
6. Perroni F, Fittipaldi S, Falcioni L, Ghizzoni L, Borrione P, Vetrano M, et al. Effect of pre-season training phase on anthropometric, hormonal and fitness parameters in young soccer players. PLoS ONE. 2019;14:e0225471. https://doi.org/10.1371/journal.pone.0225471.
7. Mujika I, Padilla S. Detraining: loss of training-induced physiological and performance adaptations. Part I. Sport Med. 2000;30:79–87. https://doi.org/10.2165/00007256-200030020-00002.
8. Suarez-Arrones L, Lara-Lopez P, Maldonado R, Torreno N, De Hoyo M, Yuzo Nakamura F, et al. The effects of detraining and retraining periods on fat-mass and fat-free mass in elite male soccer players. PeerJ. 2019;7:e7466. https://doi.org/10.7717/peerj.7466.
9. Requena B, García I, Suárez-Arrones L, De Villarreal ES, Naranjo Orellana J, Santalla A. Off-season effects on functional performance, body composition, and blood parameters in top-level professional soccer players. J Strength Cond Res. 2017;31:939–46. https://doi.org/10.1519/JSC.0000000000001568.
10. Rodriguez-Fernandez A, Sanchez-Sanchez J, Ramirez-Campillo R, Rodriguez-Marroyo JA, Vicente JGV, Yuzo NF. Effects of short-term in-season break detraining on repeated-sprint ability and intermittent endurance according to initial performance of soccer player. PLoS ONE. 2018;13:e201111. https://doi.org/10.1371/journal.pone.0201111.
11. Koundourakis NE, Androulakis NE, Malliaraki N, Margioris AN. Vitamin D and exercise performance in professional soccer players. PLoS ONE. 2014;9:e101659. https://doi.org/10.1371/journal.pone.0101659.
12. Vassilis S, Yiannis M, Athanasios M, Dimitrios M, Ioannis G, Thomas M. Effect of a 4-week detraining period followed by a 4-week strength program on isokinetic strength in elite youth soccer players. J Exerc Rehabil. 2019;15:67–73.
13. Slettaløkken G, Rønnestad BR. High-intensity interval training every second week maintains VO_{2max} in soccer players during off-season. J Strength Cond Res. 2014;28:1946–51. https://doi.org/10.1519/JSC.0000000000000356.
14. Sotiropoulos A, Travlos AK, Gissis I, Souglis AG, Grezios A. The effect of a 4-week training regimen on body fat and aerobic capacity of professional soccer players during the transition period. J Strength Cond Res. 2009;23:1697–703. https://doi.org/10.1519/JSC.0b013e3181b3df69.
15. Heidt RS, Sweeterman LM, Carlonas RL, Traub JA, Tekulve FX. Avoidance of soccer injuries with preseason conditioning. Am J Sports Med. 2000;28:659–62.
16. Coppalle S, Rave G, Abderrahman AB, Ali A, Salhi I, Zouita S, et al. Relationship of pre-season training load with in-season biochemical markers, injuries and performance in professional soccer players. Front Physiol. 2019;10:409. https://doi.org/10.3389/fphys.2019.00409.
17. Jaspers A, Kuyvenhoven JP, Staes F, Frencken WGP, Helsen WF, Brink MS. Examination of the external and internal load indicators' association with overuse injuries in professional soccer players. J Sci Med Sport. 2018;21:579–85. https://doi.org/10.1016/j.jsams.2017.10.005.
18. Malone S, Owen A, Mendes B, Hughes B, Collins K, Gabbett TJ. High-speed running and sprinting as an injury risk factor in soccer: can well-developed physical qualities reduce the risk? J Sci Med Sport. 2018;21:257–62. https://doi.org/10.1016/j.jsams.2017.05.016.
19. Huijgen BCH, Elferink-Gemser MT, Lemmink KAPM, Visscher C. Multidimensional performance characteristics in selected and deselected talented soccer players. Eur J Sport Sci. 2014;14:2–10. https://doi.org/10.1080/17461391.2012.725102.
20. Stølen T, Chamari K, Castagna C, Wisløff U. Physiology of soccer: an update. Sport Med. 2005;35:501–36.
21. Taylor JB, Wright AA, Dischiavi SL, Townsend MA, Marmon AR. Activity demands during multi-directional team sports: a systematic review. Sport Med. 2017;47:2533–51. https://doi.org/10.1007/s40279-017-0772-5.
22. Turner AN, Stewart PF. Strength and conditioning for soccer players. Strength Cond J. 2014;36:1–13. https://doi.org/10.1519/SSC.0000000000000054.
23. Redkva PE, Paes MR, Fernandez R, Da-Silva SG. Correlation between match performance and field tests in professional soccer players. J Hum Kinet. 2018;62:213–9. https://doi.org/10.1515/hukin-2017-0171.
24. Carling C, Orhant E. Variation in body composition in professional soccer players: interseasonal and intraseasonal changes and the effects of exposure time and player position. J Sports Med Phys Fit. 2010;24:1332–9.
25. Green S, Higgins J. Cochrane handbook for systematic reviews of interventions. Hoboken: Wiley; 2005.
26. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med. 2009;6:e1000097. https://doi.org/10.1371/journal.pmed.1000097.
27. Cochrane Consumers and communication. Data Extraction Template for Included Studies. 2016. Access at https://cccrg.cochrane.org/sites/cccrg.cochrane.org/files/public/uploads/det_2015_revised_final_june_20_2016_nov_29_revised.doc.
28. Vandenbroucke JP, von Elm E, Altman DG, Gøtzsche PC, Mulrow CD, Pocock SJ, et al. Strengthening the reporting of observational studies in epidemiology (STROBE): explanation and elaboration. PLoS Med. 2007;4:e297. https://doi.org/10.1371/journal.pmed.0040297.
29. Skrede T, Steene-Johannessen J, Anderssen SA, Resaland GK, Ekelund U. The prospective association between objectively measured sedentary time, moderate-to-vigorous physical activity and cardiometabolic risk factors in youth: a systematic review and meta-analysis. Obes Rev. 2019;20:55–74. https://doi.org/10.1111/obr.12758.
30. García-Hermoso A, Ramírez-Campillo R, Izquierdo M. Is muscular fitness associated with future health benefits in children and adolescents? A systematic review and meta-analysis of

longitudinal studies. Sport Med. 2019;49:1079–94. https://doi.org/10.1007/s40279-019-01098-6.
31. Moran J, Ramirez-Campillo R, Granacher U. Effects of jumping exercise on muscular power in older adults: a meta-analysis. Sport Med. 2018;48:2843–57. https://doi.org/10.1007/s40279-018-1002-5.
32. Deeks JJ, Higgins JP, Altman DG. Analysing data and undertaking meta-analyses. In: Higgins JP, Green S, editors. Cochrane handbook for systematic reviews of intervention. London: The Cochrane Collaboration; 2008. p. 243–96.
33. Kontopantelis E, Springate DA, Reeves D. A re-analysis of the cochrane library data: the dangers of unobserved heterogeneity in meta-analyses. PLoS ONE. 2013;8:e69930. https://doi.org/10.1371/journal.pone.0069930.
34. Hardy RJ, Thompson SG. A likehood approach to meta-analysis with random effects. Stat Med. 1996;15:619–29. https://doi.org/10.1002/(SICI)1097-0258(19960330)15:6%3c619::AID-SIM188%3e3.0.CO;2-A.
35. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. Med Sci Sport Exerc. 2009;41:3–13.
36. Higgins JP, Deeks JJ, Altman DG. Special topics in statistics. In: Higgings JP, Green S, editors. Cochrane handbook for systematic review of interventions. London: The Cochrane Collaboration; 2008. p. 481–529.
37. Khoury B, Lecomte T, Fortin G, Masse M, Therien P, Bouchard V, et al. Mindfulness-based therapy: a comprehensive meta-analysis. Clin Psychol Rev. 2013;33:763–71.
38. Hofmann SG, Wu JQ, Boettcher H. Effect of cognitive–behavioral therapy for anxiety disorders on quality of life: a meta-analysis. J Consult Clin Psychol. 2014;82:375–91.
39. Higgins JPT. Measuring inconsistency in meta-analyses. BMJ. 2003;327:557–60. https://doi.org/10.1136/bmj.327.7414.557.
40. Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gøtzsche PC, Ioannidis JP, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. Ann Intern Med. 2009;151:65–94. https://doi.org/10.1016/j.jclinepi.2009.06.006.
41. Higgins JPT, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med. 2002;21:1539–58. https://doi.org/10.1002/sim.1186.
42. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. BMJ. 1997;315:629–34. https://doi.org/10.1136/bmj.315.7109.629.
43. Christensen PM, Krustrup P, Gunnarsson TP, Kiilerich K, Nybo L, Bangsbo J. VO_2 kinetics and performance in soccer players after intense training and inactivity. Med Sci Sport Exerc. 2011;43:1716–24. https://doi.org/10.1249/mss.0b013e318211c01a.
44. Thomassen M, Christensen PM, Gunnarsson TP, Nybo L, Bangsbo J. Effect of 2-wk intensified training and inactivity on muscle Na +–K+ pump expression, phospholemman (FXYDI) phosphorylation, and performance in soccer players. J Appl Physiol. 2010;108:898–905. https://doi.org/10.1152/japplphysiol.01015.2009.
45. Reinke S, Karhausen T, Doehner W, Taylor W, Hottenrott K, Duda GN, et al. The influence of recovery and training phases on body composition, peripheral vascular function and immune system of professional soccer players. PLoS ONE. 2009;4:e4910. https://doi.org/10.1371/journal.pone.0004910.
46. Rossi FE, Landreth A, Beam S, Jones T, Norton L, Cholewa JM. The effects of a sports nutrition education intervention on nutritional status, sport nutrition knowledge, body composition, and performance during off season training in NCAA Division I baseball players. J Sports Sci Med. 2017;16:60–8.
47. Melchiorri G, Ronconi M, Triossi T, Viero V, De Sanctis D, Tancredi V, et al. Detraining in young soccer players. J Sport Med Phys Fit. 2014;54:27–33.
48. Caldwell BP, Peters DM. Seasonal variation in physiological fitness of a semiprofessional soccer team. J Strength Cond Res. 2009;23:1370–7. https://doi.org/10.1519/JSC.0b013e3181b3df69.
49. Koundourakis NE, Androulakis NE, Malliaraki N, Tsatsanis C, Venihaki M, Margioris AN. Discrepancy between exercise performance, body composition, and sex steroid response after a six-week detraining period in professional soccer players. PLoS ONE. 2014;9:e87803. https://doi.org/10.1371/journal.pone.0087803.
50. Cavar M, Marsic T, Corluka M, Culjak Z, Cerkez Zovko I, Müller A, et al. Effects of 6 weeks of different high-intensity interval and moderate continuous training on aerobic and anaerobic performance. J Strength Cond Res. 2019;33:44–56. https://doi.org/10.1519/JSC.0000000000002798.
51. Joo CH. The effects of short term detraining and retraining on physical fitness in elite soccer players. PLoS ONE. 2018;13:0196212. https://doi.org/10.1371/journal.pone.0196212.
52. Nakamura D, Suzuki T, Yasumatsu M, Akimoto T. Moderate running and plyometric training during off-season did not show a significant difference on soccer-related high-intensity performance compared with no-training controls. J Strength Cond Res. 2012;26:3392–7. https://doi.org/10.1519/JSC.0b013e3182474356.
53. Ramírez-Campillo R, Gallardo F, Henriquez-Olguín C, Meylan CMP, Martínez C, Álvarez C, et al. Effect of vertical, horizontal, and combined plyometric training on explosive, balance, and endurance performance of young soccer players. J Strength Cond Res. 2015;29:1784–95. https://doi.org/10.1519/JSC.0000000000000827.
54. Ramirez-Campillo R, Álvarez C, García-Hermoso A, Ramírez-Vélez R, Gentil P, Asadi A, et al. Methodological characteristics and future directions for plyometric jump training research: a scoping review. Sport Med. 2018;48:1059–81. https://doi.org/10.1007/s40279-018-0870-z.
55. Maio Alves JMV, Rebelo AN, Abrantes C, Sampaio J. Short-term effects of complex and contrast training in soccer players' vertical jump, sprint, and agility abilities. J Strength Cond Res. 2010;24:936–41. https://doi.org/10.1519/JSC.0b013e3181c7c5fd.
56. Haugen T, Tønnessen E, Hisdal J, Seiler S. The role and development of sprinting speed in soccer. Int J Sports Physiol Perform. 2014;9:432–41. https://doi.org/10.1123/ijspp.2013-0121.
57. Rumpf MC, Lockie RG, Cronin JB, Jalilvand F. Effect of different sprint training methods on sprint performance over various distances. J Strength Cond Res. 2016;30:1767–85. https://doi.org/10.1519/JSC.0000000000001245.
58. Haugen TA, Tønnessen E, Seiler S. Anaerobic performance testing of professional soccer players 1995–2010. Int J Sports Physiol Perform. 2013;8:148–56. https://doi.org/10.1123/ijspp.8.2.148.
59. Spencer M, Bishop D, Dawson B, Goodman C. Physiological and metabolic responses of repeated-sprint activities. Sport Med. 2005;35:1025–44. https://doi.org/10.2165/00007256-200535120-00003.
60. Buchheit M, Mendez-Villanueva A, Delhomel G, Brughelli M, Ahmaidi S. Improving repeated sprint ability in young elite soccer players: repeated shuttle sprints vs. explosive strength training. J Strength Cond Res. 2010;24:2715–22. https://doi.org/10.1519/JSC.0b013e3181bf0223.
61. Solberg Nedrehagen E, Hole SA. The effects of in-season repeated sprint training compared to regular soccer training. J Hum Kinet. 2015;49:237–44.
62. Halson SL, Lastella M. Amazing athletes with ordinary habits: why is changing behavior so difficult? Int J Sports Physiol Perform. 2017;12:1273–4. https://doi.org/10.1123/ijspp.2017-0632.

Affiliations

Filipe Manuel Clemente[1,2] · Rodrigo Ramirez-Campillo[3,4] · Hugo Sarmento[5]

1 Escola Superior Desporto e Lazer, Instituto Politécnico de Viana do Castelo, Rua Escola Industrial e Comercial de Nun'Álvares, 4900-347 Viana do Castelo, Portugal

2 Delegação da Covilhã, Instituto de Telecomunicações, 1049-001 Lisboa, Portugal

3 Human Performance Laboratory, Quality of Life and Wellness Research Group, Department of Physical Activity Sciences, Universidad de Los Lagos, Lord Cochrane 1046, Osorno, Chile

4 Centro de Investigación en Fisiología del Ejercicio, Facultad de Ciencias, Universidad Mayor, Santiago, Chile

5 Research Unit for Sport and Physical Activity, Faculty of Sport Sciences and Physical Education, University of Coimbra, Coimbra, Portugal